CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 7: WRITE THE PYTHON POGRAM TO IMPLEMENT BFS

Aim:

To write a Python program to implement Breadth-First Search (BFS) traversal of a graph.

Algorithm:

1. Start the program.
2. Create a graph using an adjacency list.
3. Initialize a queue and a visited set.
4. Insert the starting node into the queue and mark it as visited.
5. Remove a node from the queue and display it.
6. Add all unvisited adjacent nodes to the queue and mark them as visited.
7. Repeat steps 5 and 6 until the queue becomes empty.
8. Stop the program.

PY 7.py - C:/Users/navit/PY 7.py (3.12.10)

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```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    visited.add(start)

    print("BFS Traversal:")

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbor in graph[node]:
            if neighbor not in visited:
                visited.add(neighbor)
                queue.append(neighbor)

graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'F'],
    'D': ['B'],
    'E': ['B', 'F'],
    'F': ['C', 'E']
}

bfs(graph, 'A')
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/PY 7.py ==========
BFS Traversal:
A B C D E F
>>>
```

Result:

The Python program was executed successfully, and the Breadth-First Search traversal was obtained.